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import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, classification_report,
    roc_auc_score, roc_curve
)
data = load_breast_cancer()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target, name="target")
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42,
    stratify=y)
rf = RandomForestClassifier(random_state=42, n_jobs=-1)
param_grid = {
    "n_estimators": [100],
    "max_depth": [None, 10],
    "min_samples_split": [2],
    "min_samples_leaf": [1]}
grid = GridSearchCV(
    estimator=rf,
    param_grid=param_grid,
    scoring="f1",
    cv=3,
    n_jobs=1,
    verbose=1
)
grid.fit(X_train, y_train)
print("Best Parameters (CV):", grid.best_params_)
best_rf = grid.best_estimator_
best_rf.fit(X_train, y_train)
y_pred = best_rf.predict(X_test)
y_prob = best_rf.predict_proba(X_test)[:, 1]
acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred, zero_division=0)

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f1 = f1_score(y_test, y_pred)
auc = roc_auc_score(y_test, y_prob)
cm = confusion_matrix(y_test, y_pred)
print("\n=== Random Forest – Test Metrics ===")
print(f"Accuracy : {acc:.4f}")
print(f"Precision: {prec:.4f}")
print(f"Recall : {rec:.4f}")
print(f"F1-Score : {f1:.4f}")
print(f"ROC-AUC : {auc:.4f}")
print("\nConfusion Matrix:\n", cm)
print("\nClassification Report:\n", classification_report(y_test, y_pred, zero_division=0))
importances = pd.Series(best_rf.feature_importances_, index=X.columns)
top10 = importances.sort_values(ascending=False).head(10)
plt.figure()
top10[::1].plot(kind="barh")
plt.xlabel("Importance")
plt.title("Top 10 Feature Importances – Random Forest")
plt.grid(axis="x", alpha=0.3)
plt.show()
fpr, tpr, thresholds = roc_curve(y_test, y_prob)
plt.figure()
plt.plot(fpr, tpr, label=f"Random Forest (AUC = {auc:.3f})")
plt.plot([0, 1], [0, 1], linestyle="--", color='gray')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve – Random Forest")
plt.legend()
plt.grid(True)
plt.show()

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Fitting 3 folds for each of 2 candidates, totalling 6 fits
Best Parameters (CV): {'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}

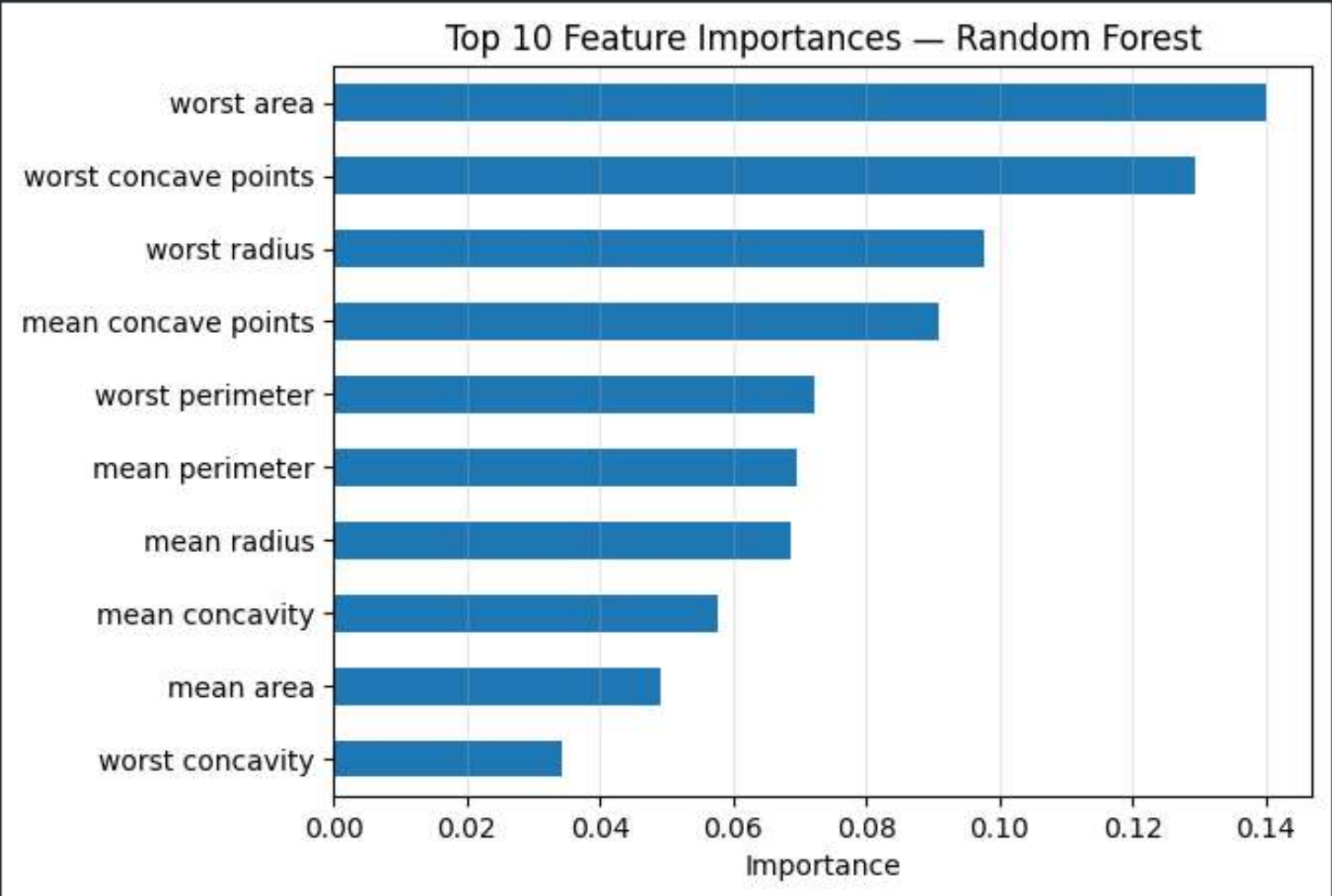
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=== Random Forest – Test Metrics ===
Accuracy : 0.9561
Precision: 0.9589

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Classification Report:					
		precision	recall	f1-score	support
	0	0.95	0.93	0.94	42
	1	0.96	0.97	0.97	72
accuracy				0.96	114
macro avg		0.96	0.95	0.95	114
weighted avg		0.96	0.96	0.96	114



Importance

ROC Curve — Random Forest

